

SCOPE : L0 — Survival

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Chapter 0: Introduction

The Sephirotic names used in this paper are not intended in a religious or mystical sense. In SCOPE, they are borrowed as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for estimating cognitive structures in humans and AI. Here, "coordinate system" does not mean a flowchart implying a fixed sequence of cognitive processing, but rather a framework for organizing and estimating cognitive structures. Each layer does not represent a hierarchy or a temporal order.

In SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is only the output itself; it is not possible to directly know what cognitive structure operated internally to produce it. For this reason, SCOPE uses observable output as a clue to estimate the cognitive mechanism behind it.

This paper addresses L0, one such cognitive structure. L0 is the layer responsible for survival and the maintenance of life. However, L0 is not defined as the starting point of cognition.

SCOPE does not assume that cognition begins from any specific layer; rather, it treats cognition as a structure in which each layer mutually influences the others.

This paper examines L0's characteristics, its circuit relationships with other layers, and the correspondence of roles between humans and AI, centered on this role as a survival foundation. Note that some points, such as the fixation of physical skills, remain at a hypothetical stage and are left as topics for future examination.

Chapter 1: What L0 Is

L0 is not instinct itself, but a layer responsible for a set of cognitive functions that operate to sustain survival. Instinctive needs are treated as one part of this.

It handles instinctive needs required to stay alive, such as hunger, sleep, and danger avoidance. It also governs responses that maintain survival in reaction to changes in the external environment or within the body.

L0, as used here, does not refer to biological organs or the nervous system itself. In SCOPE, it is positioned as a cognitive layer for organizing and estimating the functions related to survival and the maintenance of life.

Furthermore, L0 is not defined as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any specific layer; rather, it treats cognition as a structure in which each layer mutually influences the others. L0, therefore, does not exist independently of other layers, but functions as part of the overall cognitive structure.

This paper positions L0 as a survival foundation and proceeds to organize its characteristics and its relationships with other layers.

Chapter 2: Basic Characteristics of L0

Chapter 1 defined L0 as the layer responsible for the cognitive functions that sustain survival and the maintenance of life. This chapter, building on that definition, organizes the basic characteristics of L0.

Unlike other layers, L0 operates continuously. Regardless of one's state of thought or emotion, the functions related to survival and the maintenance of life continue without interruption; a state in which L0 fully stops is not assumed within this model.

L0 also carries strong priority in situations involving survival. For example, in life-threatening situations such as a fire or a fall, responses related to danger avoidance take precedence and can strongly affect other cognitive activity.

However, this priority does not imply a fixed hierarchy. SCOPE does not define any predetermined ranking among layers. L0 is prioritized only in situations where survival and the maintenance of life are strongly demanded.

In summary, L0 operates continuously and exerts strong influence in survival-related situations. At the same time, no fixed ranking among layers is defined, and L0 is not positioned as a predetermined starting point within cognitive processing as a whole. How L0 relates to other layers will be addressed in Chapter 4.

Chapter 3: L0 Changes

Chapter 2 organized the basic characteristics of L0. This chapter examines the possibility that L0 is affected by experience.

Instinct is generally understood as innate and unchanging. However, experiences strongly related to survival and the maintenance of life can have long-term effects on subsequent responses.

A representative example is trauma and PTSD. After a strong experience involving danger to life, survival responses may arise preferentially in response to similar stimuli. This kind of change may potentially be explained as an update to L0.

On the other hand, this paper does not address the acquisition of physical skills such as swimming or riding a bicycle. These phenomena do not have a clear, direct relationship to survival and the maintenance of life, and their correspondence with L0 cannot be determined at this stage. In addition, since this series primarily treats linguistic output, such as conversation and text, as its object of observation, the examination of non-linguistic physical skills currently falls outside its scope. These points are left as topics for future examination.

It is also known that similar responses can arise not only from experiences involving danger to life, but also from ongoing psychological pressure or an oppressive environment. Whether such phenomena can be explained as changes in L0, or should instead be understood through interaction with other layers such as L2 or L3, cannot be determined at this stage and is left as a topic for future examination.

For these reasons, this paper treats it as a hypothesis that strong experiences — including not only physical danger but also psychological pressure or an oppressive environment — may affect L0.

Chapter 4: L0 Interacts with Other Layers

Chapter 3 organized the possibility that L0 changes under the influence of experience. This chapter examines the possibility that such change arises through interaction with other layers.

L0 does not function independently. Since cognitive processing operates as a circuit in which multiple layers influence one another, L0 is also considered to change within its relationships with other layers.

For example, in cases of physical danger such as a traffic accident, a survival response arises, and this experience may affect L0.

On the other hand, in cases of ongoing psychological pressure or an oppressive environment, effects may extend not only to emotions such as fear and anxiety, but also to self-perception and interpersonal cognition. As a result, change may reach L0 through interaction with other layers, potentially leading to a physical survival response in similar situations thereafter.

At present, it cannot be determined whether change in L0 originates from a specific layer, or whether L0 is updated as a result of simultaneous change across multiple layers.

However, it can at least be said that change in L0 does not necessarily occur in isolation, and may arise through interaction with other layers.

For this reason, SCOPE treats L0 not as an isolated function, but as part of a circuit involving other layers.

Chapter 5: The Relationship Between L0 and Malkuth

Chapter 4 organized the possibility that L0 changes through interaction with other layers. This chapter examines the relationship between L0 and Malkuth.

L0 itself cannot be directly observed. What can be observed is the result of survival-related cognitive processing manifesting in reality.

For example, sweating, trembling, goosebumps, increased heart rate, flight behavior, and screaming may be observed as instances of this.

However, these phenomena alone do not allow one to directly determine the involvement of L0. As discussed in Chapter 4, L0 does not function in isolation but changes through interaction with other layers; therefore, even the same physical response may involve multiple layers.

For this reason, SCOPE does not judge the state of L0 directly from observed output alone, but instead uses the results that appear in Malkuth as a clue to infer the cognitive structure behind them.

This point applies not only to L0 but to other layers as well. What SCOPE handles is always an inference based on observable output, not direct access to the layer itself.

Chapter 6: Correspondence with LLMs

Chapter 5 organized the relationship between L0 and Malkuth. This chapter examines the correspondence between L0 and LLMs.

SCOPE is a coordinate system for organizing and estimating human cognitive structures. Accordingly, it is not intended to map each human layer directly onto LLMs.

At the same time, there may be processes in LLMs that play functionally similar roles to those in humans.

In humans, L0 is responsible for cognition related to survival and the maintenance of life. LLMs, however, have no biological instinct or survival function. Therefore, one cannot assume that LLMs possess a function identical to human L0.

At present, a process in LLMs that temporarily holds input information and passes it on to subsequent processing — something like a staging buffer — may correspond functionally to L0.

However, this correspondence is a hypothesis based on functional similarity, and does not imply that the internal processing of human L0 and that of LLMs are identical.

It should also be noted that simple input-output state transition mechanisms are not considered within the scope of SCOPE's application. SCOPE addresses cognitive structures of a certain complexity, in which multiple processes may be involved with one another.

SCOPE does not treat humans and LLMs as identical structures, but rather organizes and estimates each of their cognitive structures using the same coordinate system. Examining their functional correspondence makes it possible to consider each cognitive structure from a shared perspective.

Conclusion

This paper organized L0 as a cognitive layer related to survival and the maintenance of life.

L0 is not instinct itself, but a layer responsible for the cognition that sustains survival and the maintenance of life. It operates continuously and exerts strong influence in survival-related situations, while not being positioned as a fixed hierarchy or a fixed starting point.

It was also examined that L0 may change through experiences involving not only physical danger but also psychological pressure or an oppressive environment, and that such change may arise not in isolation but through interaction with other layers.

Furthermore, this paper organized SCOPE's basic method of observation: L0 itself cannot be directly observed, and the cognitive structure behind it is inferred using the output that appears in Malkuth as a clue.

Regarding correspondence with LLMs, this was positioned as a hypothesis based on functional similarity at present, showing that humans and LLMs are not treated as identical structures.

The content presented in this paper is an organization based on current observation, and includes some hypotheses that require further examination. It may be revised through future observation and verification.

The next paper will address L1 (linguistic expression). L1 will organize the role and characteristics of how cognition is expressed as language.